

魏红陈

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Agentic AI | Generative Environments | Post-training

教育背景

武汉大学 博士研究生, 导师: 陈震中教授 (IIP Lab)

2023.09 – 2027.06

南京理工大学 (硕士) / 西安石油大学 (本科)

2020.09 – 2023.04 / 2016.09 – 2020.06

研究概览

研究目标: 面向需要整合信息、调用工具并产出实际交付物的专业工作, 研究通用智能体的能力提升, 重点探索生成式环境、数据合成、模型后训练与可验证评测。

当前重点: 牵头开展 OfficeTown 研究, 构建生成式组织模拟框架, 通过持续组织活动产生任务、工作材料与反馈, 为智能体训练与评测提供可扩展资源。

方法与评测: 研究可变状态交互、主动记忆、环境内强化学习、模型能力迁移与自对齐, 并构建证据级评测体系, 诊断复杂任务中的能力边界与失败原因。

研究与实习经历

微软亚洲研究院 (MSRA) Visual Computing Group

北京

实习生; Mentor: Dr. Bei Liu

2025.12 – 至今

OfficeTown | Generative Organizations for Evolving Agent Intelligence

ICLR 2027 在投

- 牵头开展 OfficeTown 研究, 与合作者共同构建生成式组织模拟框架, 通过持续组织活动生成专业工作任务、配套材料与反馈, 为智能体评测、监督微调和强化学习提供数据及可执行环境。
- 从自然语言业务描述生成专业领域的组织、专业角色、共享知识、客户关系与工具工作空间。构建覆盖 8 个专业领域的 23 个组织, 建立任务与评分规则均经领域专家审核的专业工作基准; 评测多种前沿模型, 揭示其在专业工作任务上的能力缺口。使用合成数据训练 Qwen3.5-9B, 在包括 GDPval/Jobbench 在内的专业工作基准上取得明显提升; 下游 Harness 对照实验表明, principal-assistant 架构优于所测试的单智能体与多智能体基线。

DocAtlas | 可变状态文档交互与环境内强化学习

NeurIPS 2026

- 牵头开展 DocAtlas 研究, 与合作者将长文档理解从静态检索问答重构为可变状态交互; 设计统一 Search、Read、Note 与 Review 的可训练文档 Harness, 结合自改进检索与主动记忆, 支持大模型推理及强化学习。
- 在 MMLongBench-Doc 上, GPT-5.4 配合 DocAtlas 达到 71.4%, 高于人类专家参考结果 (65.8%); Qwen3.5-4B 经环境内 RL 达到 63.7%, 较 54.4% 基线提升 9.3 个百分点。

XL-DocBench | 超长文档的证据级评测与失败诊断

NeurIPS 2026

- 牵头开展 XL-DocBench 研究, 与合作者共同构建结合专家标注证据页、类型化验证规则与 12 类推理标签的评测体系, 将最终答案评分扩展至可审计的证据级诊断, 区分检索、证据使用与规则执行错误。
- 基准经 194 位领域专家完整验证, 覆盖 6 个专业领域、1,519 道问题及最长 2,303 页上下文, 包含多页、跨文档、多模态与不可回答问题; 进一步构建约 40K 合成后训练样本。

武汉大学 IIP 实验室

武汉

研究方向: 模型能力迁移、自对齐与长上下文理解

2023.04 – 至今

模型能力迁移与自对齐 | PASA、ANO

- 牵头开展 PASA 研究, 刻画模型合并后的感知-推理不对称性, 提出基于 detached-anchor 非对称 GRPO 的自对齐框架, 在无需真值答案标签的条件下修复视觉感知退化并提升推理能力。
- 牵头开展 ANO 研究, 提出按层定制的免训练模型融合方法, 利用视觉注意力的层级差异, 在保留视觉感知能力的同时迁移语言模型的推理、代码与 GUI 相关能力, 无需额外多模态训练数据或梯度更新。

- 牵头开展长上下文理解与生成系列研究，提出面向长视频理解的 **Visual Context Window Extension** 与开放世界时空定位方法 **RealVG**，并拓展至长描述生成与无监督提示学习。

代表性论文

成果概览：已发表及在投论文共 **18** 篇，其中**第一作者 11** 篇、合作论文 7 篇。以下列出全部一作论文及精选合作论文。

第一作者论文：在投

- **Hongchen Wei**, et al. “OfficeTown: Generative Organizations for Evolving Agent Intelligence.” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “PASA: Post-Merge Perception–Reasoning Asymmetry as a Self-Alignment Signal for MLLMs.” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “Layer-Specialized Model Merging for Training-Free Cross-Modal Reasoning Transfer (ANO).” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “LongCaption: Unlocking the Power of Long Caption Generation in Large Multimodal Models.” *TIST Under review*.

第一作者论文：已发表

- **Hongchen Wei**, et al. “DocAtlas: Long-Document Understanding as Mutable-State Interaction.” *NeurIPS 2026*.
- **Hongchen Wei**, et al. “XL-DocBench: Benchmarking Evidence-Grounded Extra-Long Document Understanding.” *NeurIPS 2026*.
- **Hongchen Wei**, et al. “Visual Context Window Extension: A New Perspective for Long Video Understanding.” *ACM MM 2025*.
- **Hongchen Wei**, et al. “RealVG: Unleashing MLLMs for Training-Free Spatio-Temporal Video Grounding in the Wild.” *ACM MM 2025*.
- **Hongchen Wei**, et al. “Improving Generalization of Image Captioning with Unsupervised Prompt Learning.” *ACM TOMM 2024*.
- **Hongchen Wei**, et al. “Exploiting Cross-Modal Prediction and Relation Consistency for Semi-Supervised Image Captioning.” *IEEE TCYB 2023*.
- **Hongchen Wei**, et al. “S2OSC: A Holistic Semi-Supervised Approach for Open Set Classification.” *ACM TKDD 2022*.

合作论文：精选

- “See What We Cannot See: A Geo-guided Reasoning Benchmark for Object Counting under Adverse Earth Observation Conditions.” *CVPR 2026*.
- “TDSAgent: A Task-Driven Sampling Agent for Long Video Question Answering.” *IEEE TMM 2026*.
- “Remote Sensing Semantic Segmentation Quality Assessment based on Vision Language Model.” *IEEE TGRS 2025*.

技术能力

系统与工具：Python、Harness；工具调用与持久化环境。

训练与评测：PyTorch、vLLM；SFT、GRPO、环境内 RL、合成数据与评测。

Hongchen Wei

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Education

Wuhan University Ph.D. student; advisor: Prof. Zhenzhong Chen (IIP Lab)	Sep. 2023 – Jun. 2027
Nanjing University of Science and Technology M.S.	Sep. 2020 – Apr. 2023
Xi'an Shiyou University B.E.	Sep. 2016 – Jun. 2020

Research Overview

Research agenda: Generalist agents for professional work requiring information integration, tool use, and concrete deliverables, with a focus on generative environments, synthetic data, post-training, and verifiable evaluation.

Current focus: Leading OfficeTown with collaborators to generate tasks, working materials, and feedback through ongoing organizational simulation, providing scalable training and evaluation resources.

Methods and evaluation: Mutable-state interaction, active memory, in-environment RL, capability transfer, and self-alignment; evidence-level diagnosis of capability limits and failure modes.

Research Experience

Microsoft Research Asia (MSRA) Visual Computing Group	Beijing, China
<i>Intern; Mentor: Dr. Bei Liu</i>	<i>Dec. 2025 – Present</i>

OfficeTown <i>Generative Organizations</i>	<i>ICLR 2027 (under review)</i>
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- Led **OfficeTown** with collaborators to develop a generative organizational simulation framework that produces professional tasks, supporting materials, and feedback through ongoing organizational activity, providing data and executable environments for agent evaluation, supervised fine-tuning, and RL.
- Generated organizations with professional roles, shared knowledge, customer relationships, and tool-enabled workspaces from natural-language business briefs. Built **23 organizations across 8 professional domains** and a benchmark with expert-reviewed tasks and rubrics; evaluated frontier models to identify capability gaps. Training **Qwen3.5-9B** on synthetic data substantially improved **GDPval** and **Jobbench** performance. The principal–assistant harness outperformed the single- and multi-agent baselines tested.

DocAtlas <i>Mutable-State Document Interaction</i>	<i>NeurIPS 2026</i>
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- Led **DocAtlas** with collaborators, reframing long-document understanding as mutable-state interaction rather than static retrieval-based QA. Designed a trainable harness with Search, Read, Note, and Review, self-improving retrieval, and active memory for large-model inference and RL.
- On MMLongBench-Doc, GPT-5.4 + DocAtlas reached **71.4%** (human-expert reference: 65.8%). With in-environment RL, Qwen3.5-4B reached **63.7%**, versus a 54.4% baseline (**+9.3 percentage points**).

XL-DocBench <i>Evidence-Level Evaluation</i>	<i>NeurIPS 2026</i>
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- Led **XL-DocBench** with collaborators, combining expert-annotated evidence pages, typed verification rules, and **12 reasoning labels** for auditable diagnosis of retrieval, evidence-use, and rule-execution errors beyond final-answer scoring.
- Validated by **194 domain experts: 1,519 questions across 6 domains**, with contexts up to **2,303 pages**, including multi-page, cross-document, multimodal, and unanswerable questions. Further synthesized approximately **40K** post-training examples.

PASA / ANO | *Capability Transfer and Self-Alignment*

- Led **PASA** with collaborators, characterizing post-merge perception–reasoning asymmetry and developing detached-anchor asymmetric GRPO for self-alignment, recovering visual perception and improving reasoning without ground-truth answer labels.
- Led **ANO** with collaborators, developing training-free, layer-specialized model merging guided by visual-attention differences. Transferred language-model reasoning, coding, and GUI capabilities while preserving visual perception, without additional multimodal training data or gradient updates.

Long-Context Understanding and Generation

- Led studies with collaborators on **Visual Context Window Extension** for long-video understanding and **RealVG** for open-world spatio-temporal grounding, extending to long-caption generation and unsupervised prompt learning.

Selected Publications

18 papers (published or under review): **11 first-author** and 7 co-authored. Selected papers below.

First Author: Under Review

- **Hongchen Wei**, et al. “OfficeTown: Generative Organizations for Evolving Agent Intelligence.” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “PASA: Post-Merge Perception–Reasoning Asymmetry as a Self-Alignment Signal for MLLMs.” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “Layer-Specialized Model Merging for Training-Free Cross-Modal Reasoning Transfer (ANO).” *ICLR 2027 (under review)*.
- **Hongchen Wei**, et al. “LongCaption: Unlocking the Power of Long Caption Generation in Large Multimodal Models.” *ACM TIST (under review)*.

First Author: Published

- **Hongchen Wei**, et al. “DocAtlas: Long-Document Understanding as Mutable-State Interaction.” *NeurIPS 2026*.
- **Hongchen Wei**, et al. “XL-DocBench: Benchmarking Evidence-Grounded Extra-Long Document Understanding.” *NeurIPS 2026*.
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- **Hongchen Wei**, et al. “Exploiting Cross-Modal Prediction and Relation Consistency for Semi-Supervised Image Captioning.” *IEEE TCYB 2023*.
- **Hongchen Wei**, et al. “S2OSC: A Holistic Semi-Supervised Approach for Open Set Classification.” *ACM TKDD 2022*.

Selected Co-Authored Papers

- “See What We Cannot See: A Geo-guided Reasoning Benchmark for Object Counting under Adverse Earth Observation Conditions.” *CVPR 2026*.
- “TDSAgent: A Task-Driven Sampling Agent for Long Video Question Answering.” *IEEE TMM 2026*.
- “Remote Sensing Semantic Segmentation Quality Assessment based on Vision Language Model.” *IEEE TGRS 2025*.